

The Tempo and Mode of Tadpole Body Shape Evolution

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Introduction

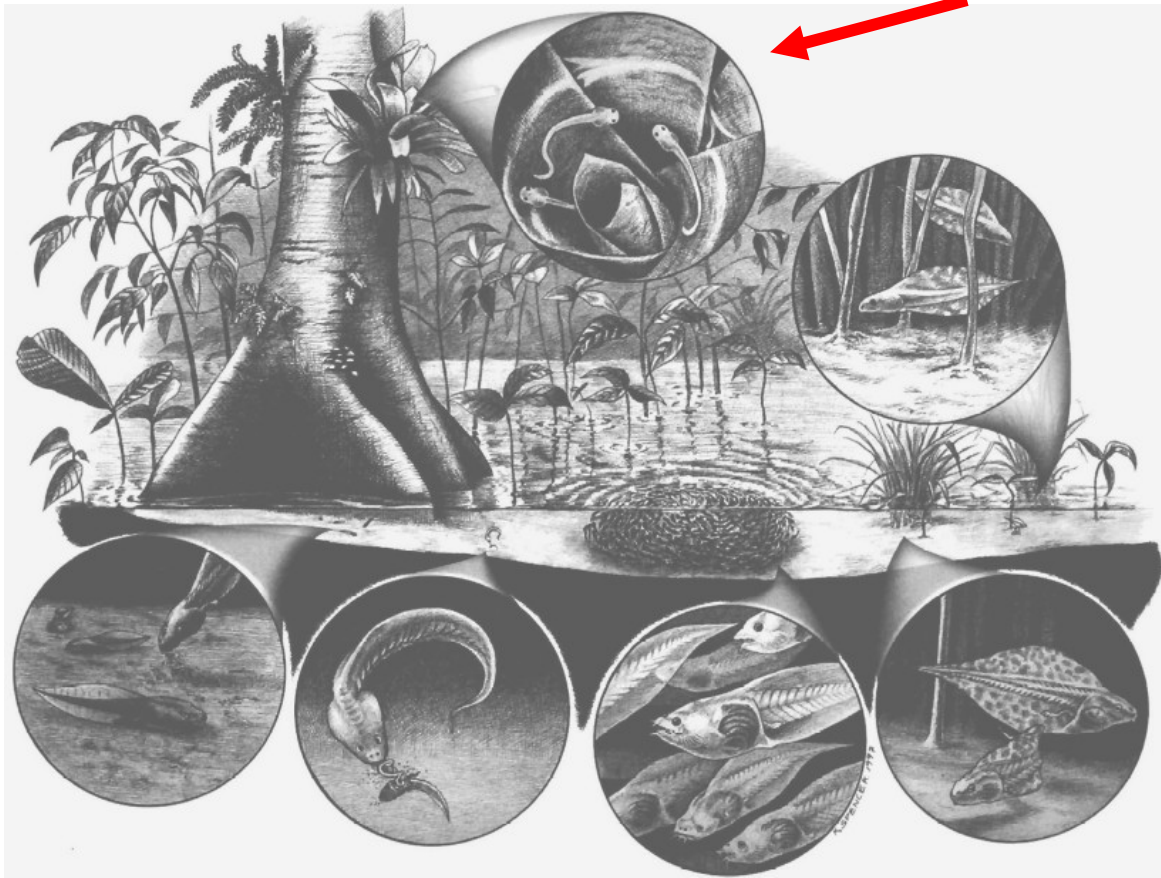


- Free-swimming tadpoles occur in habitats with distinct hydrodynamics: permanent and temporary ponds, streams, phytotelmata, and tree holes;
- Previous studies pointed out to repeated events of convergence in the morphospace of larvae throughout their evolution;
- Convergence seems related to hydrodynamics of each habitat (i.e., lentic vs. lotic);

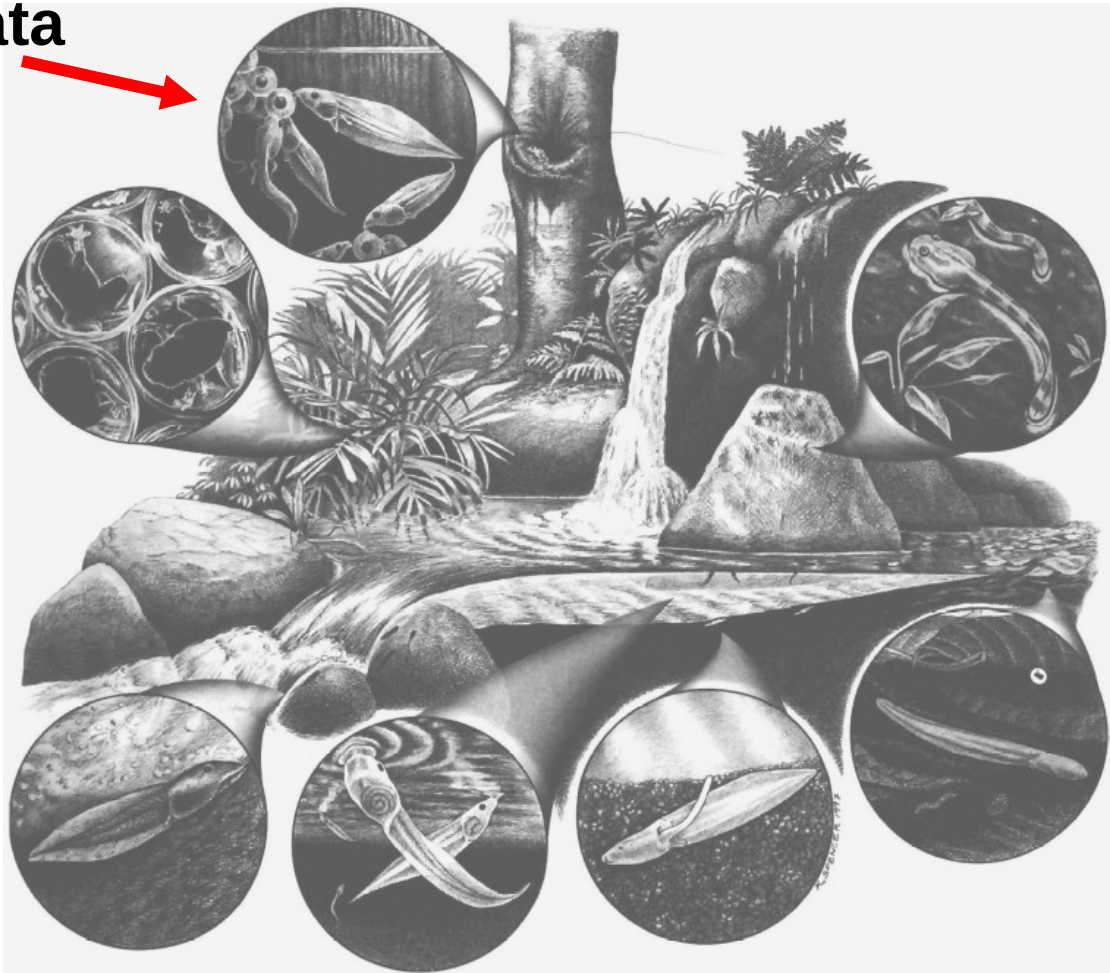
Introduction



Phytotelmata



Pond



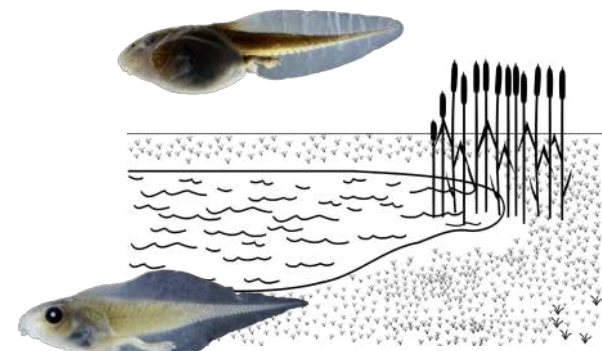
Stream

Introduction



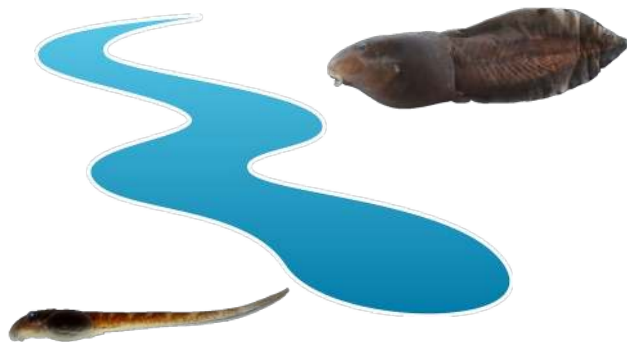
- Morphological variation in larvae is larger than adults;
- However, little is known about how macroevolutionary processes (e.g., adaptation to novel habitats) contribute to the production and maintenance of morphological variation;
- The role of habitats with contrasting micro-habitat heterogeneity (i.e., lotic vs. Lentic) on the determination of variation in tadpole body shape has not been tested yet;

Pond



- Morphological Diversity
- Ecological opportunities

Stream



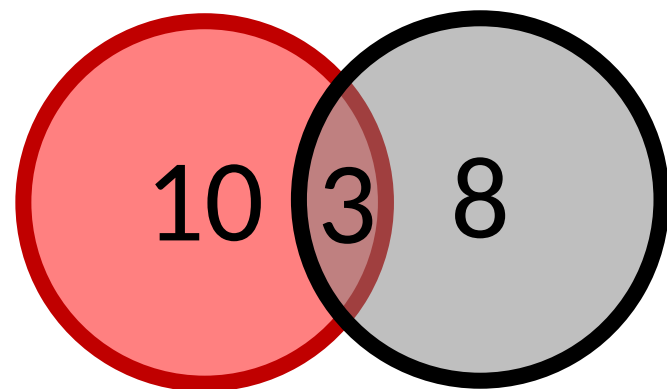
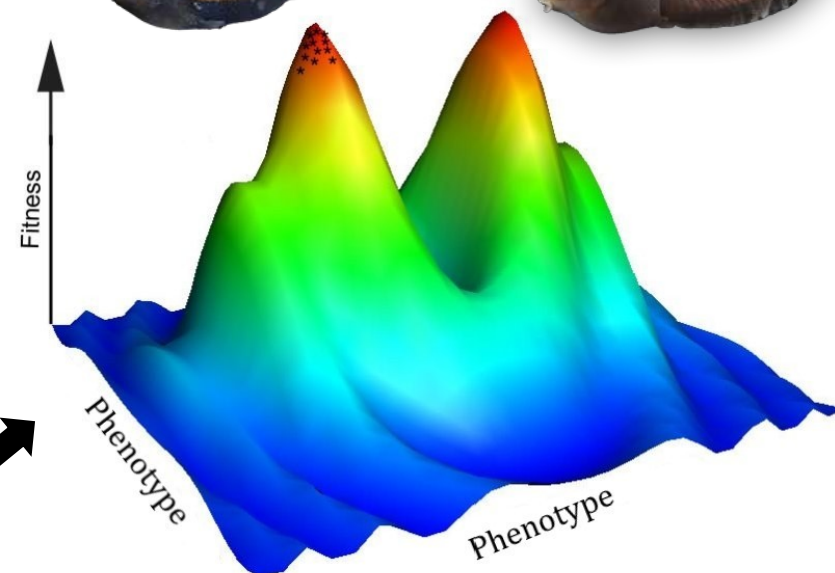
- + Morphological diversity
- + Ecological opportunities

VS.

P



R



guilds

Pattern



Lentic



Lotic

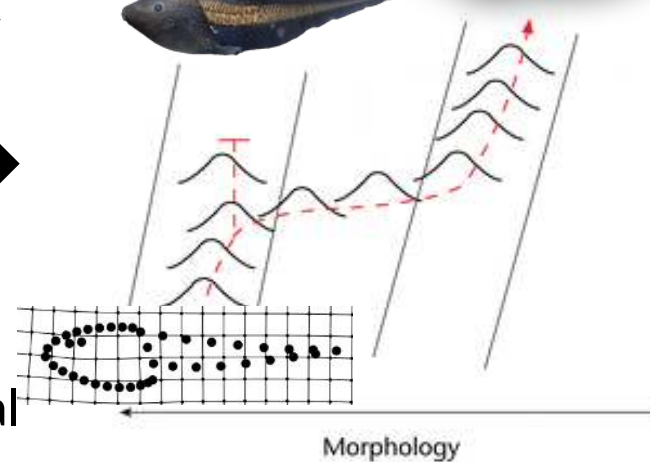


Neobatrachia

P



R



Ancestral
shape

(c)

Process

≠ Evolutionary rates
allowed lineages
to occupy
adaptive zones
and change the
adaptive
landscape

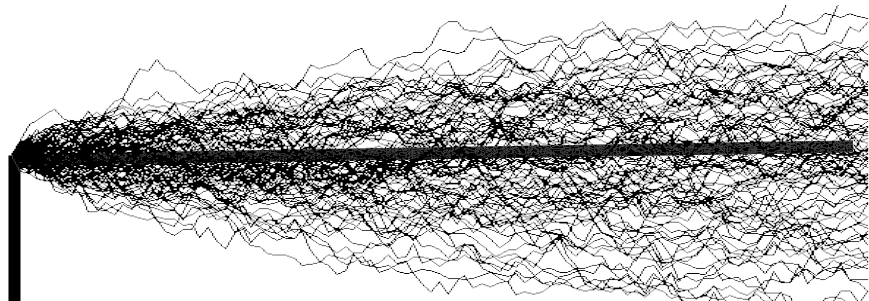
Our hypotheses

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graph TD; A[Our hypotheses] --> B((Tempo)); A --> C((Mode));
```

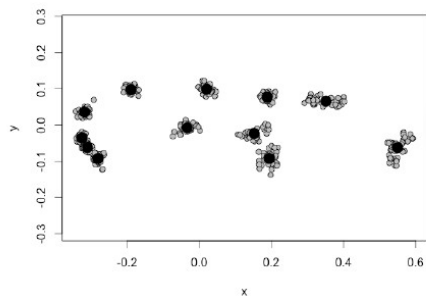
The diagram illustrates a conceptual model where a central hypothesis is divided into two distinct components. At the top, a blue rounded rectangle contains the text 'Our hypotheses'. Two blue arrows originate from the bottom of this rectangle, pointing downwards and outwards to two separate blue circles. The circle on the left is labeled 'Tempo', and the circle on the right is labeled 'Mode'. This structure suggests that the overall hypotheses are composed of or related to these two specific factors.

Tempo

Mode

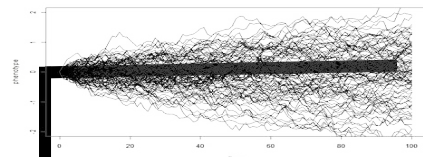


A

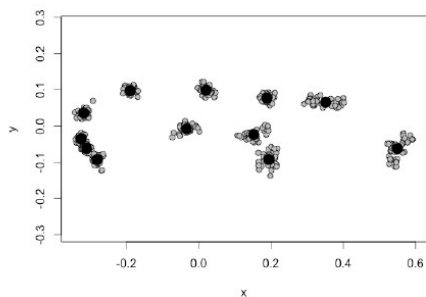


Stream

$$\sigma^2_{\text{mult_stream}}$$

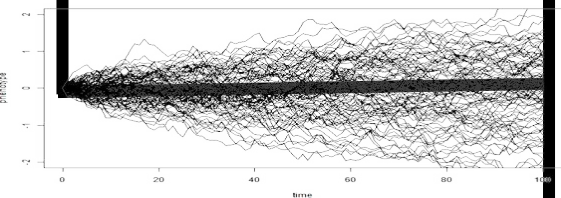


B

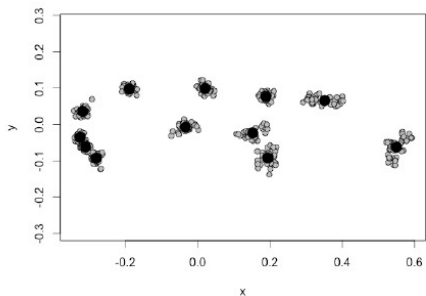
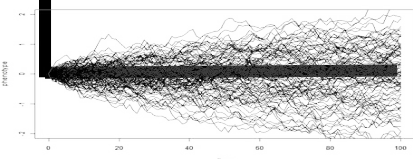


Pond

$$\sigma^2_{\text{mult_pond}}$$



C

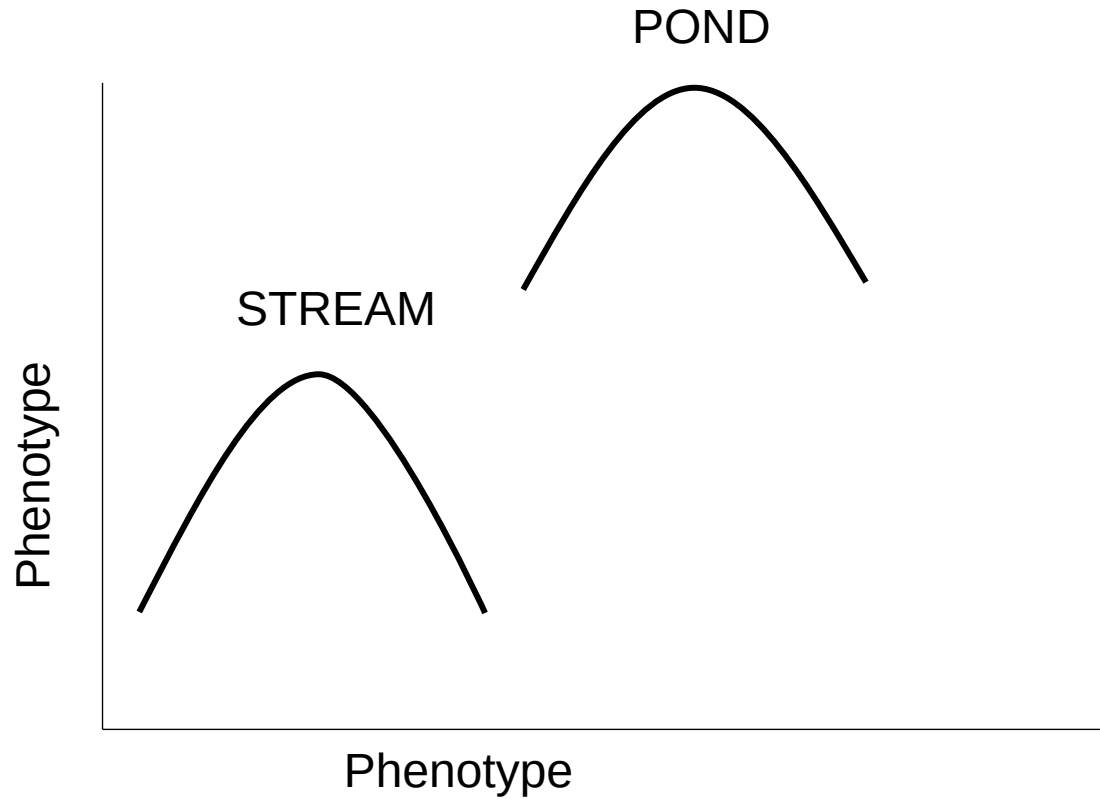


Pond

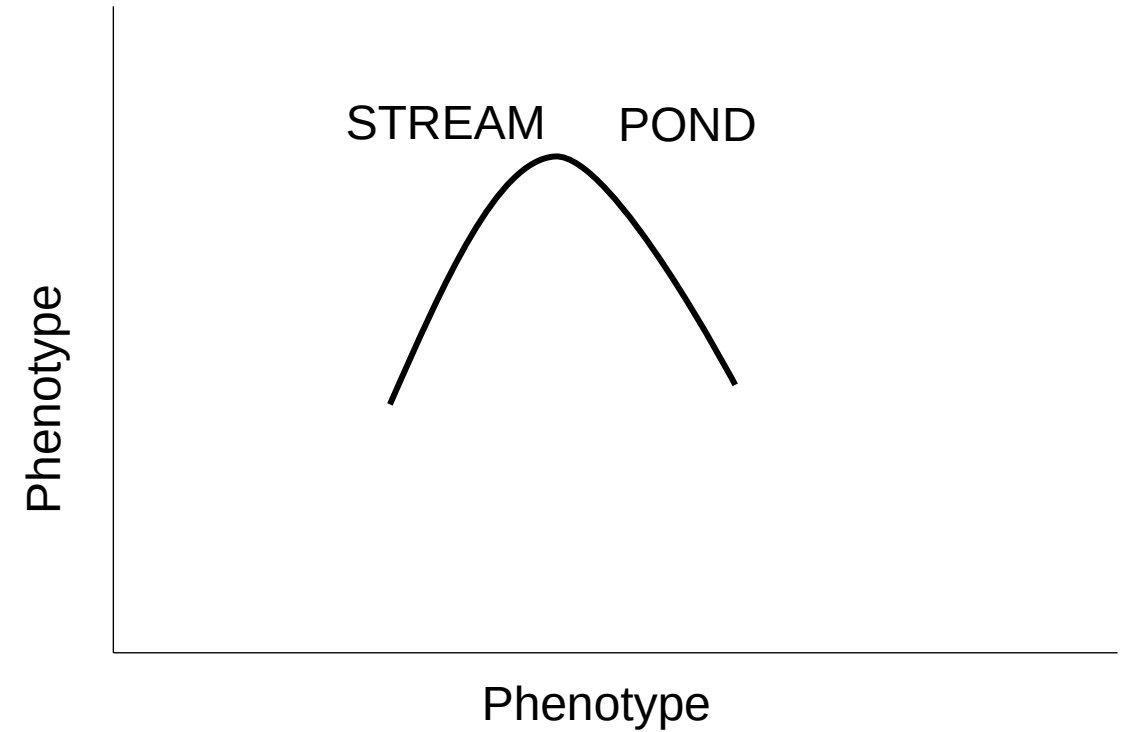
Hypothesis



Model with **two peaks** (θ)



Model with a **single peak** (θ)



Material and Methods



Geometric morphometrics

- 54 species of 11 families : Bufonidae, Centrolenidae, Ceratophrydae, Cycloramphidae, Hylidae, Hylodidae, Leptodactylidae, Microhylidae, Odontophrynidae, Ranidae and Telmatobiidae;
- Dorsal view: 11 *landmarks* and 38 *semilandmarks*:
- Lateral view 9 *landmarks* and 32 *semilandmarks*:

Habitat data:

Took from the collection's book and coded as lentic or lotic;



Landmarks



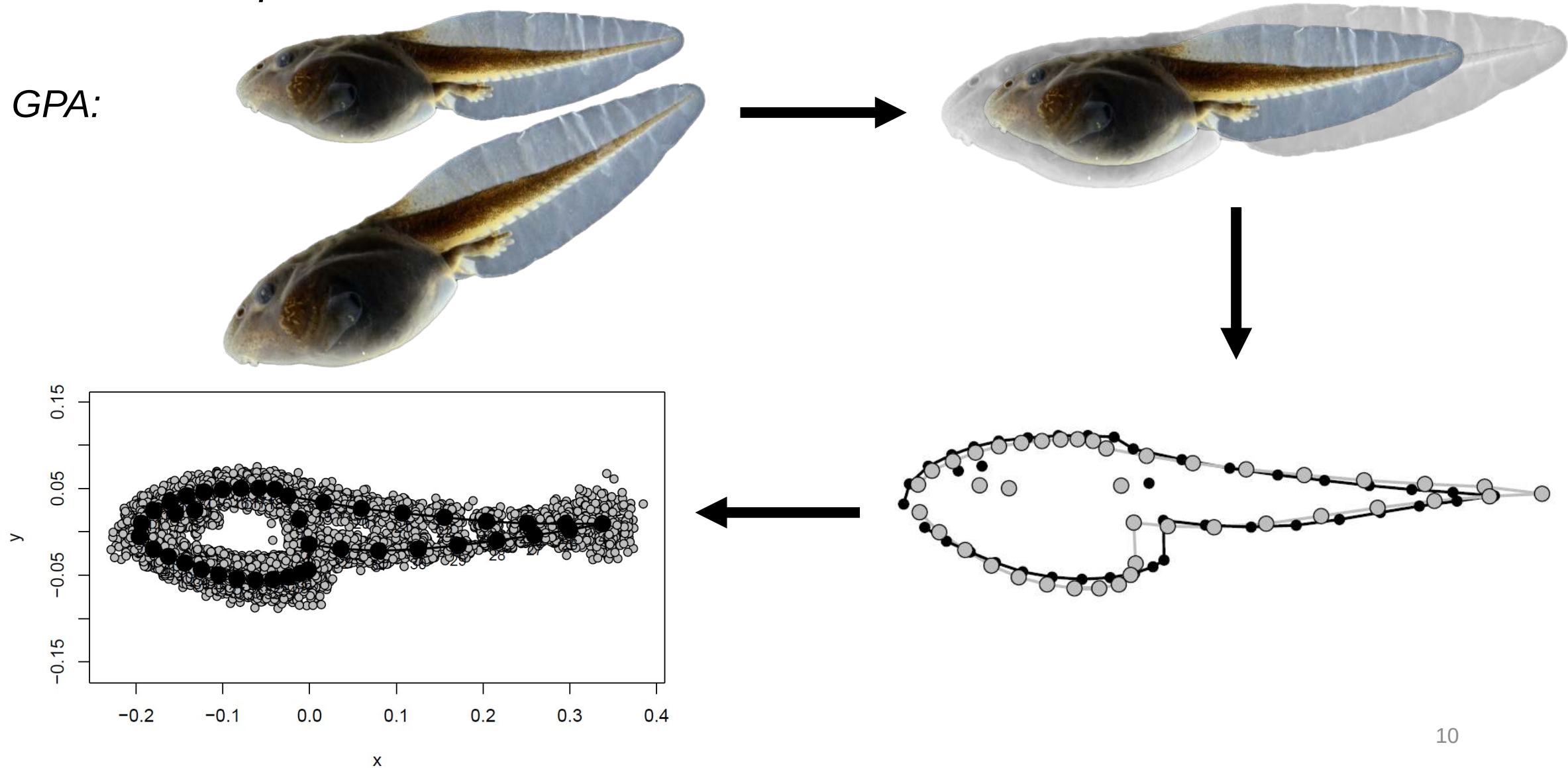
Semilandmarks



Material and Methods

Geometric morphometrics

- *GPA*:





Material and Methods

Phylogenetic Comparative Methods for multivariate data:

- *Phylomorphospace* using Jetz and Pyron 2018's tree
- We used Brownian Motion (**BM**) to estimate the rate of evolution (σ^2_{mult}) of tadpole body shape;
 - Calculates the mean rate for each discrete character (σ^2_{mult} pond, σ^2_{mult} stream) and then calculates the ratio between the two;
- Then, a Monte Carlo procedure is used to generate a null distribution of ratios, which is then compared to the observed one to obtain a *P* value;

Material and Methods



Phylogenetic Comparative Methods for multivariate data:

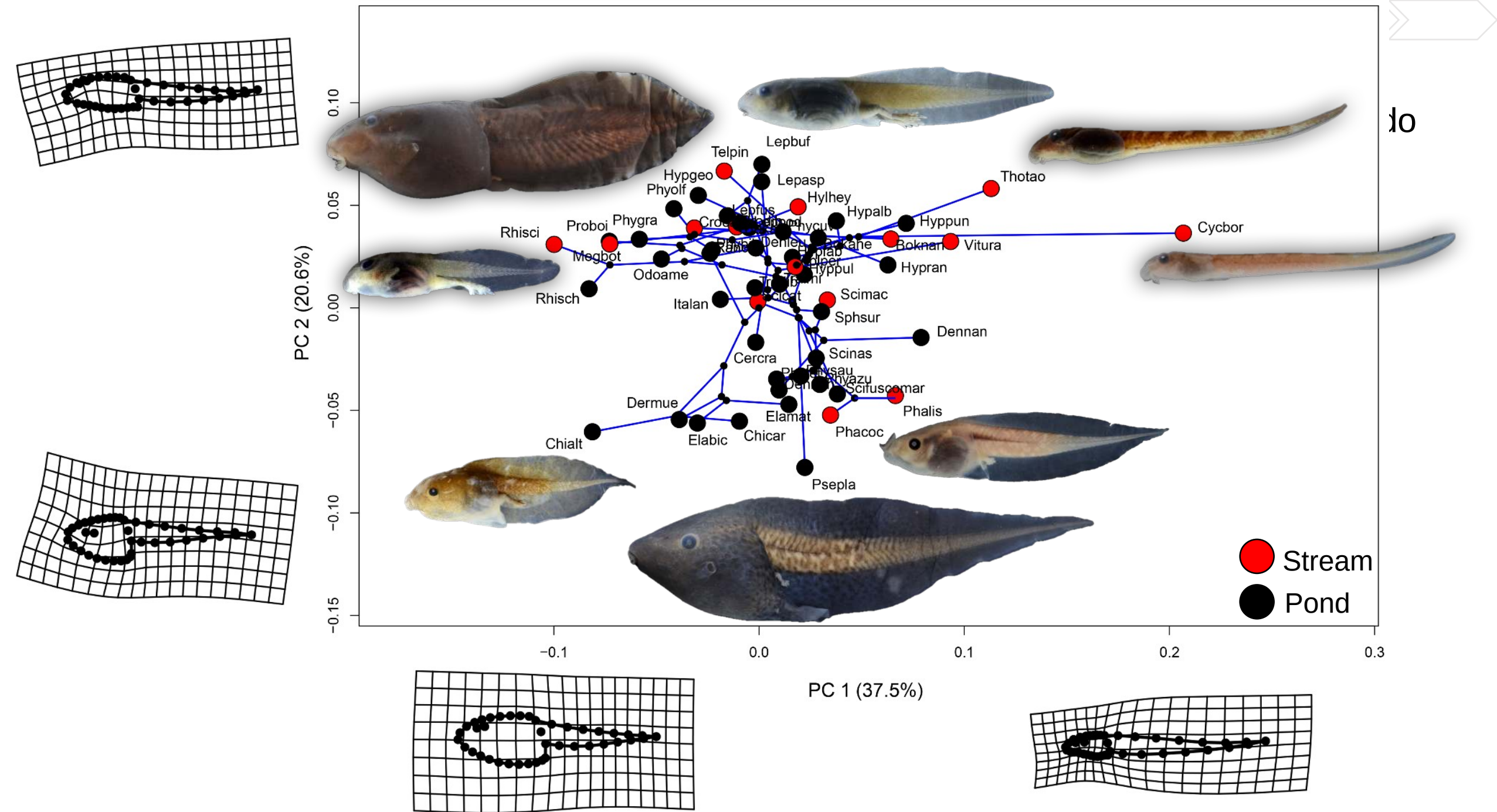
- To test if there are differences in rates of evolution of body shape **between lineages**, we used a new method that automatically finds **rate shifts** in relation to the background rate of the phylogeny or clade and **estimate the rate of evolution in each tip**;
- This analysis has been done in the package *RRphylo* 1.0.0 of *R*;



Material and Methods

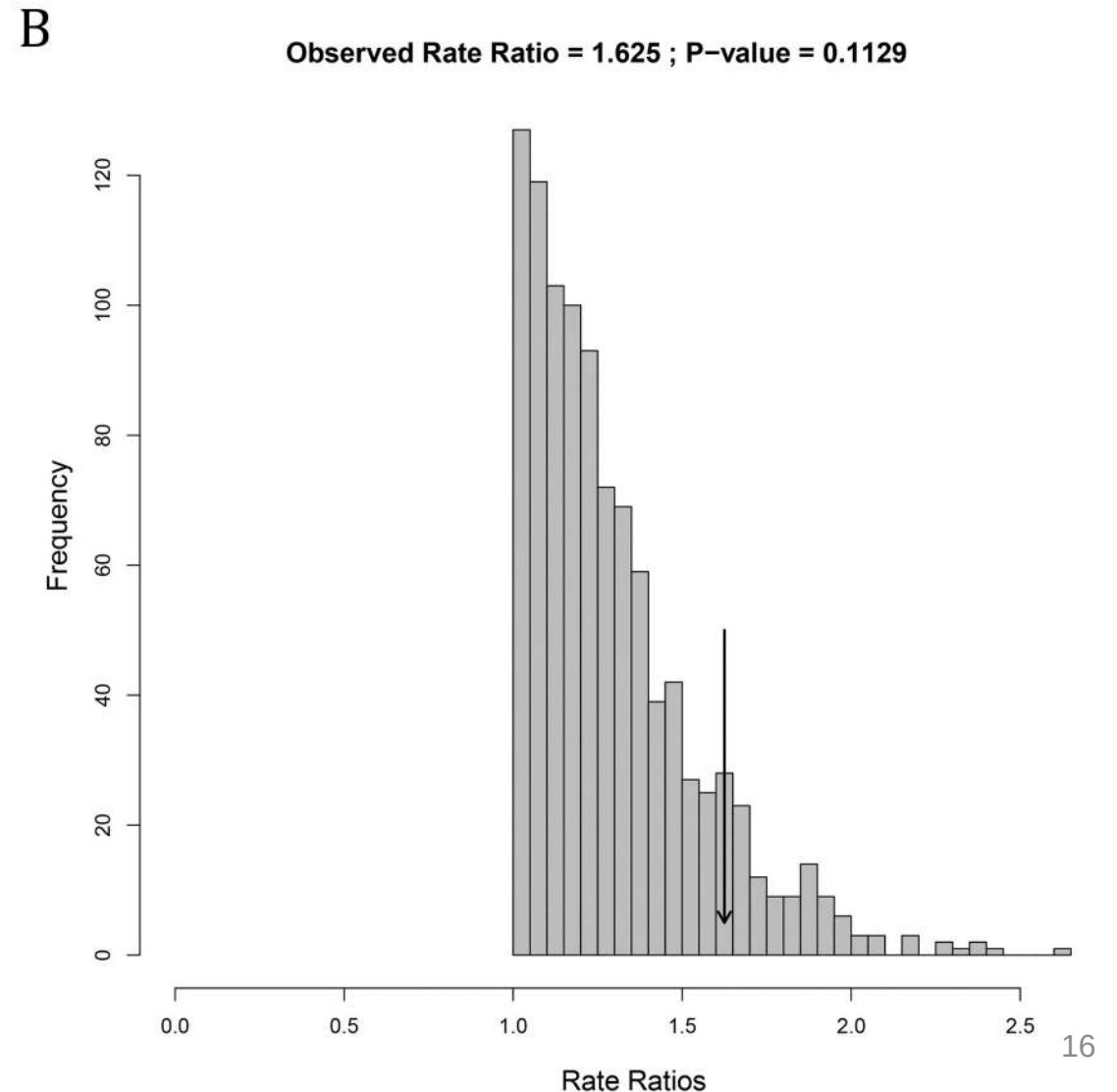
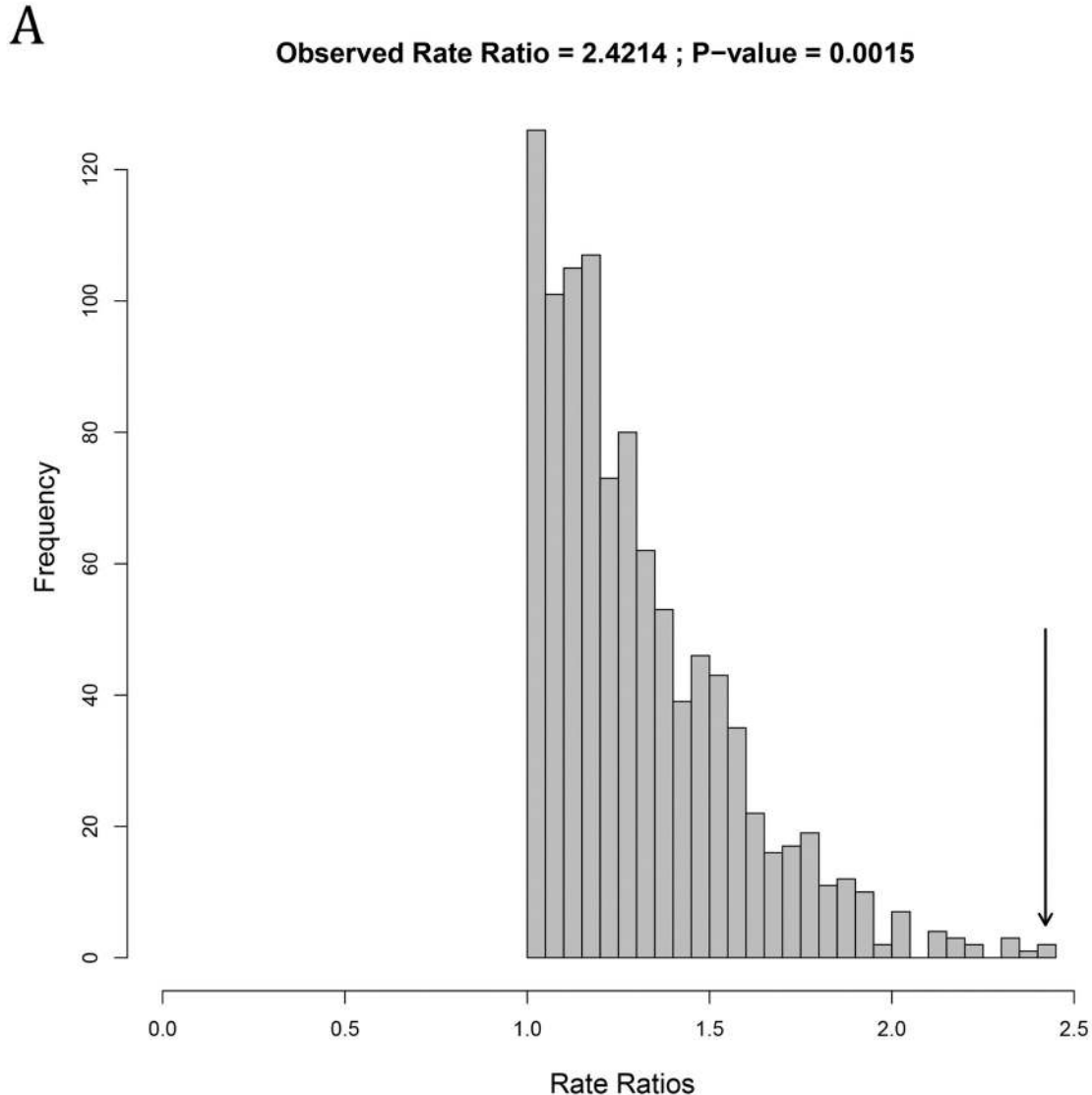
Phylogenetic Comparative Methods for multivariate data:

- To test if lineages that occupy distinct habitats belong to distinct adaptive peaks for body shape, we fitted a **OU model** with multiple peaks that detects peak shift automatically;
- For this analysis, we only used the first two PCs, selected by the Broken-stick criterion;
- This analysis was done in the package *PhylogeneticEM* 1.2.1;



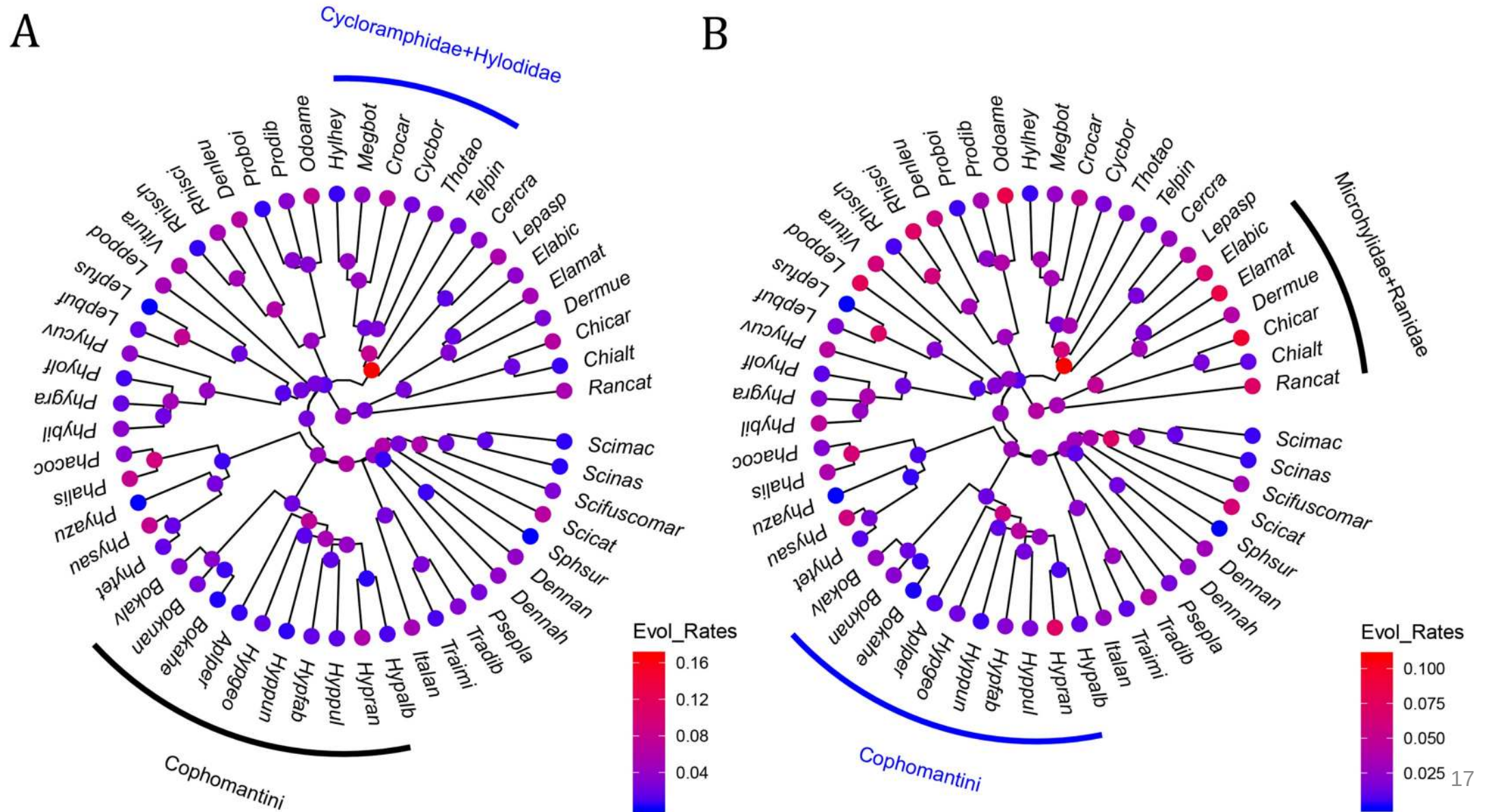
Results

The rate of evolution of body shape in lateral view for stream tadpoles ($2,1 \times 10^{-4}$) was one order of magnitude higher than that of ponds ($9,0 \times 10^{-5}$). For dorsal view, the rate of stream tadpoles ($1,0 \times 10^{-4}$) was also one order of magnitude higher than that of ponds ($6,1 \times 10^{-5}$);



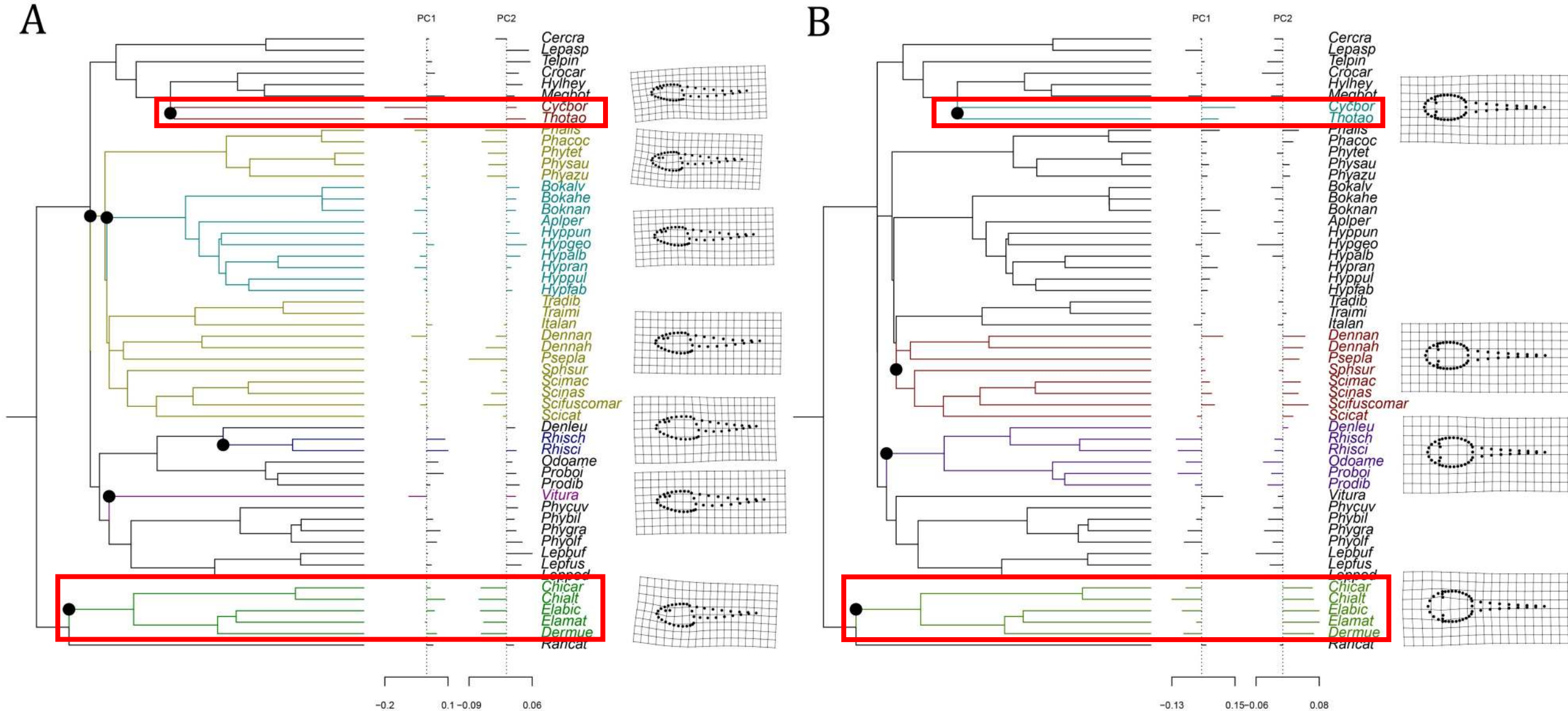
Results

We found two rate shifts for body shape for each view, dorsal and lateral;



Results

The best model to describe the adaptive landscape had six optima for lateral and four for dorsal view;

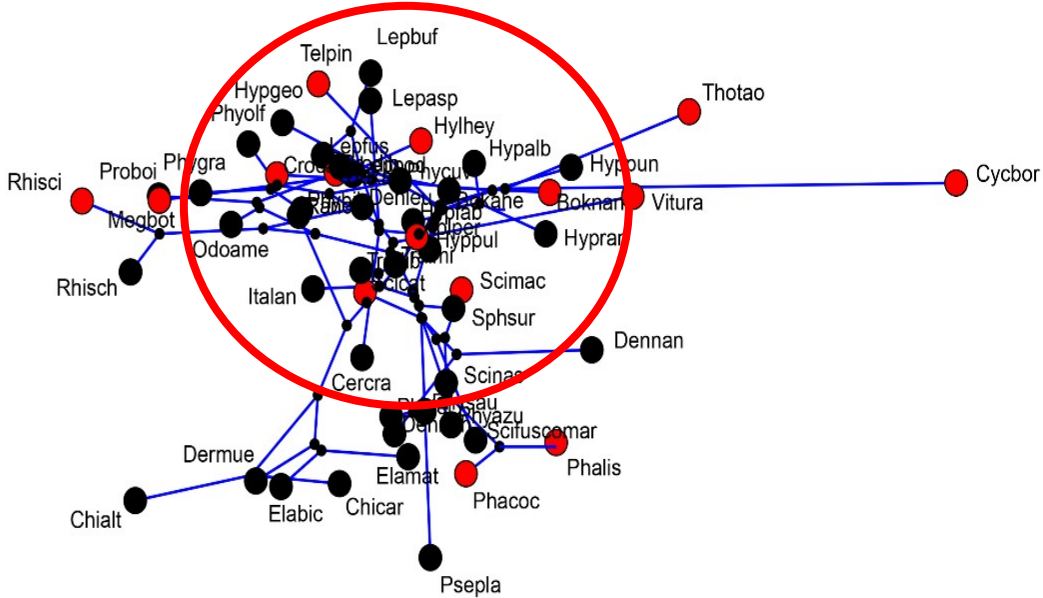




- Our results support the initial hypothesis that lineages from lentic habitats that invaded recently lotic habitats, had a different **rate of evolution** and **share distinct optima in the adaptive landscape** for body shape;
- Some lineages seem to follow a directional evolution in the morphospace, along with **clade overlap** and **disparity**;
- The **rate of evolution** of tadpole body shape from ponds is lower than that of streams in both views, but there's high heterogeneity of rate within each habitat;

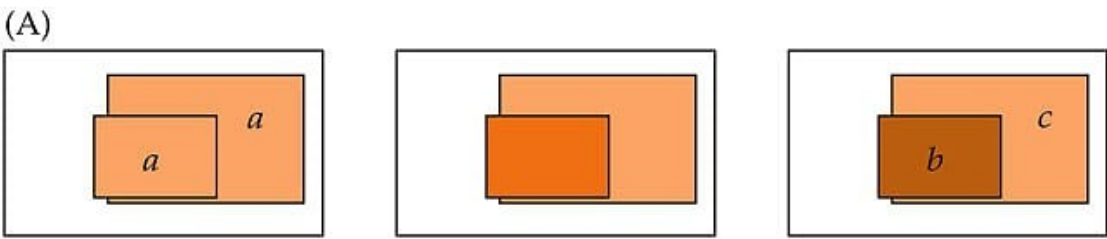
Discussion

Clade overlap > Between pond and stream species;

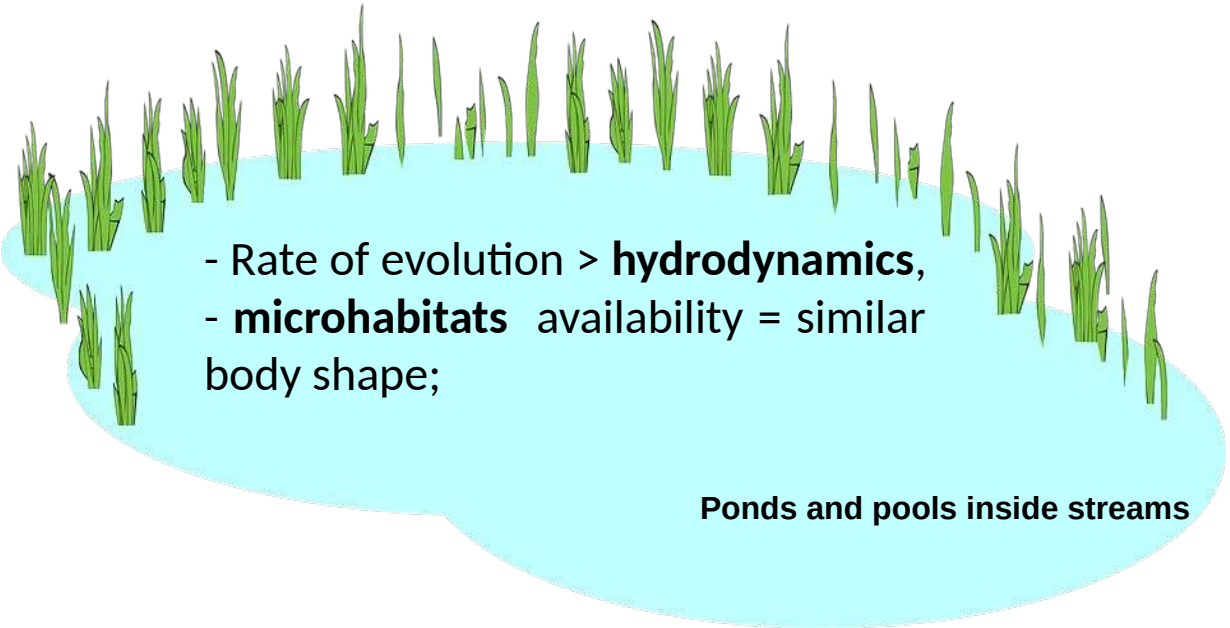


Adams and Collyer 2019

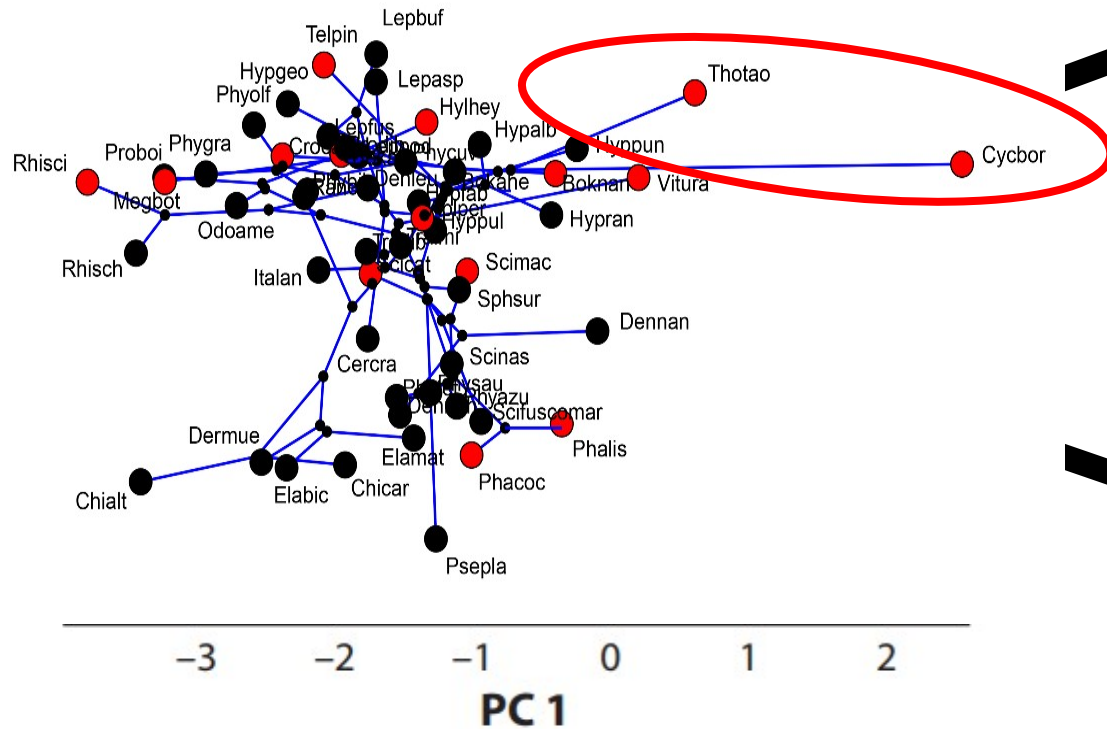
Sympatric



Lomolino et al. 2010



Directional evolution > some stream lineages occupy extrem positions in the morphospace (Cycloramphidae);

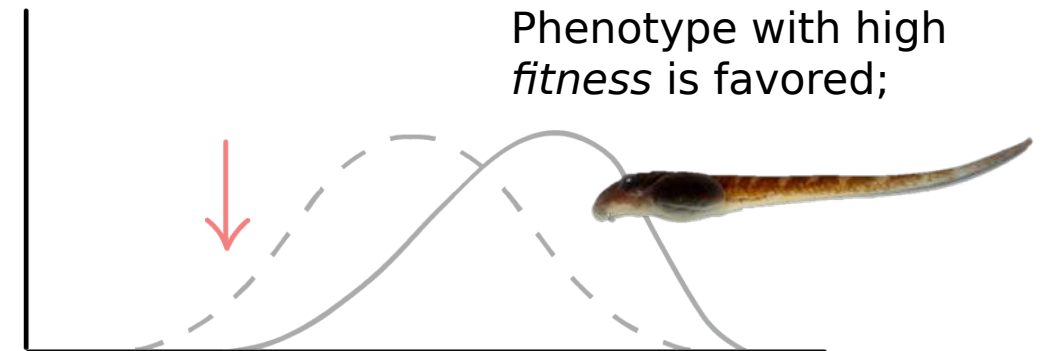


Accelerated rates of evolution >
Higher **micro-habitat availability**;
Hydrodynamics; Recent invasion of
this new habitat = **diversity of forms**;



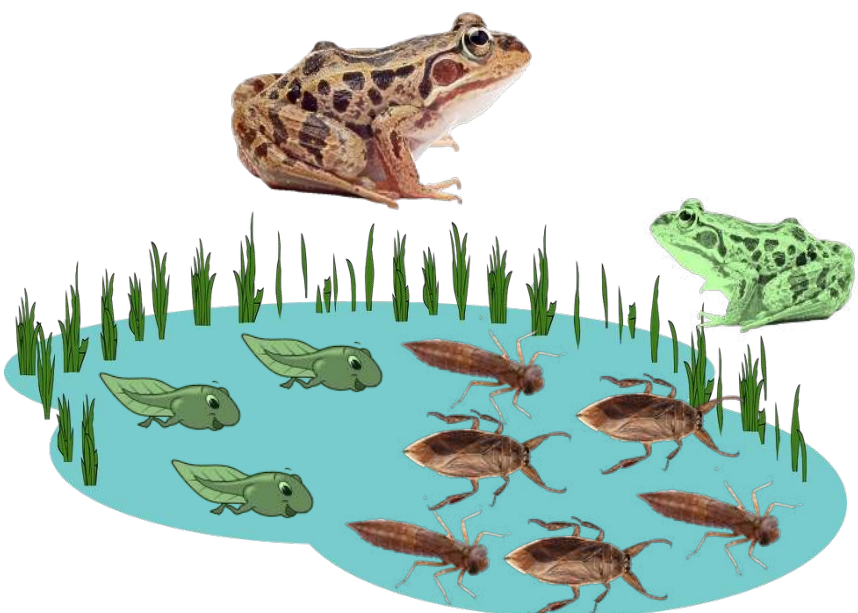
DIRECTIONAL SELECTION

Phenotype with high *fitness* is favored;

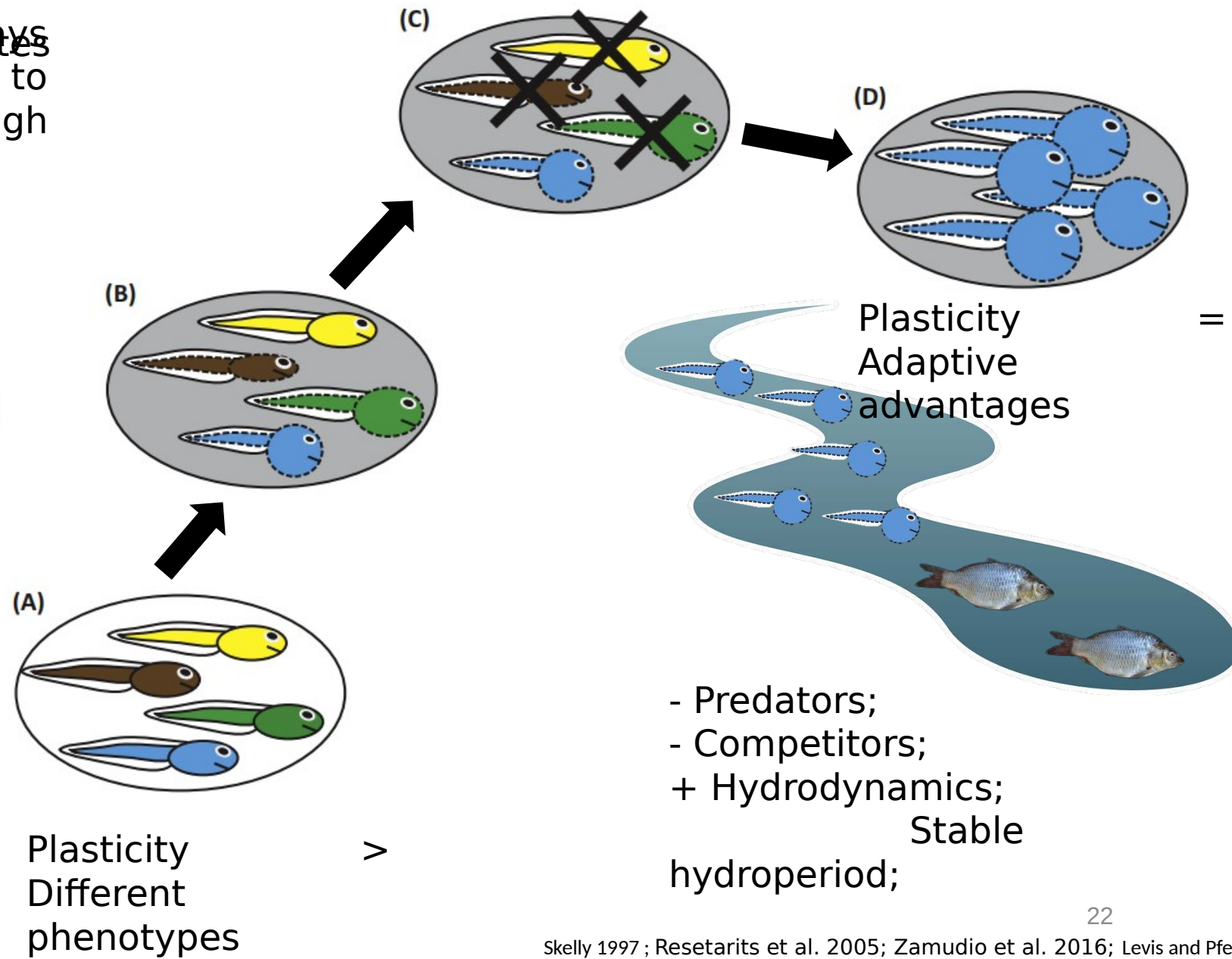


Discussion

Larvae can respond in different ways to a new development site due to **complex life cycle** and high **phenotypic plasticity**

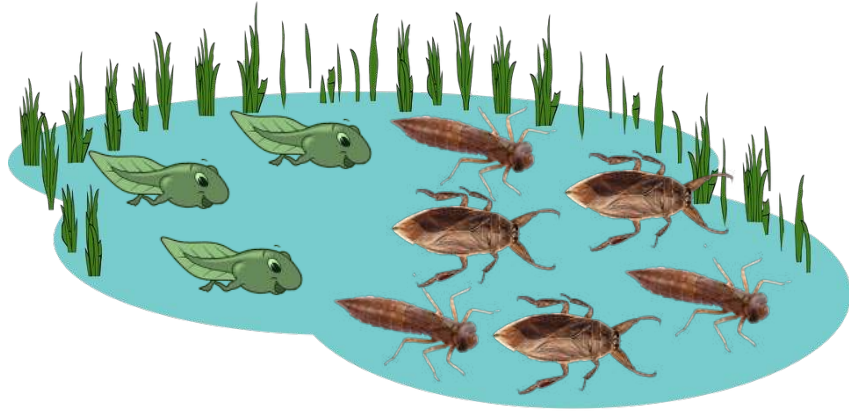


- + Predators;
 - + Competitor;
 - Hydrodynamics;
- Variable hydroperiod;



Discussion

The low primary productivity of streams would demand metamorphosis;

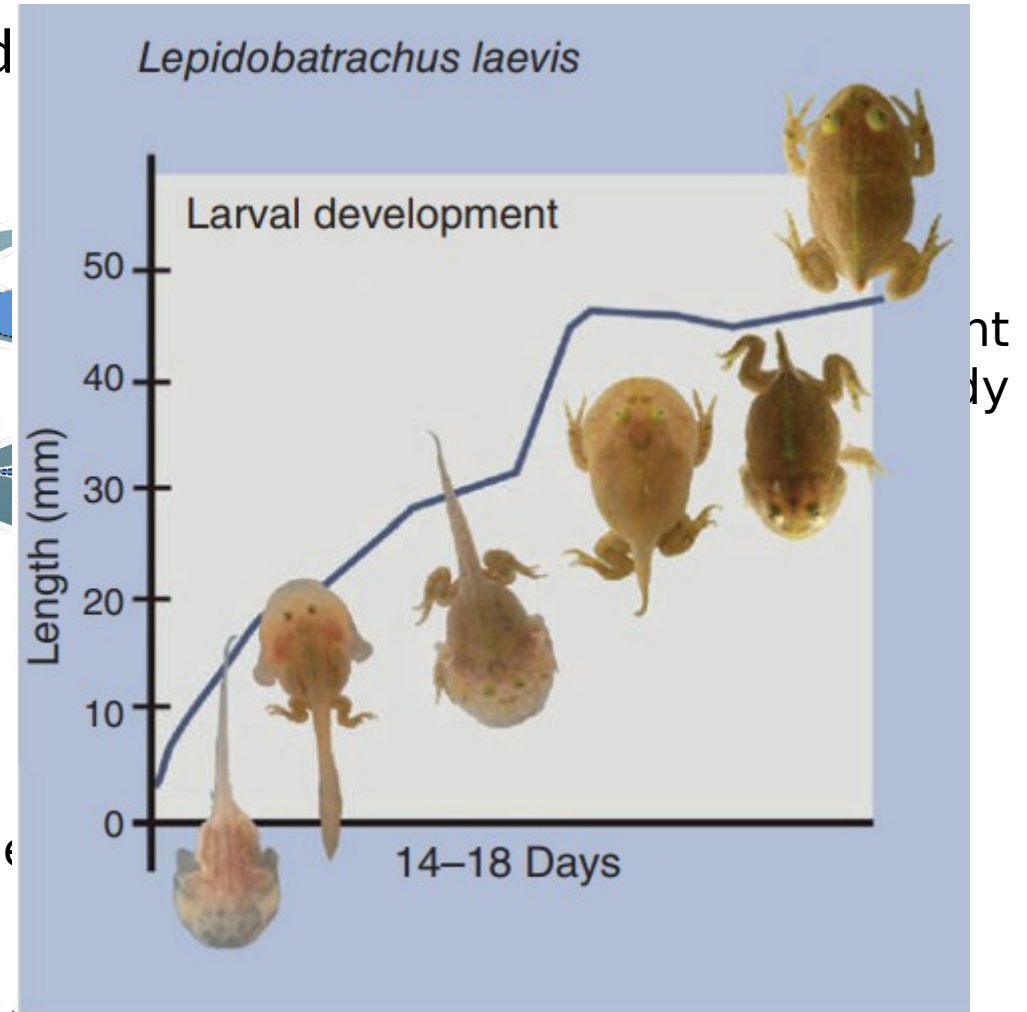


**V
S.**

> Primary productivity;
Variable hydroperiod;
Fast ontogenetic
development

This delay could have altered the timing of developmental events in which one structure is formed > evolutionary novelties;

Remain more time in the water > would imply reduced pop size, due to low juvenile recruitment |
stream frog pops = low pop size;



Fabrezi et al. 2019

Discussion

Shifting Balance Theory > stream populations = low pop size;

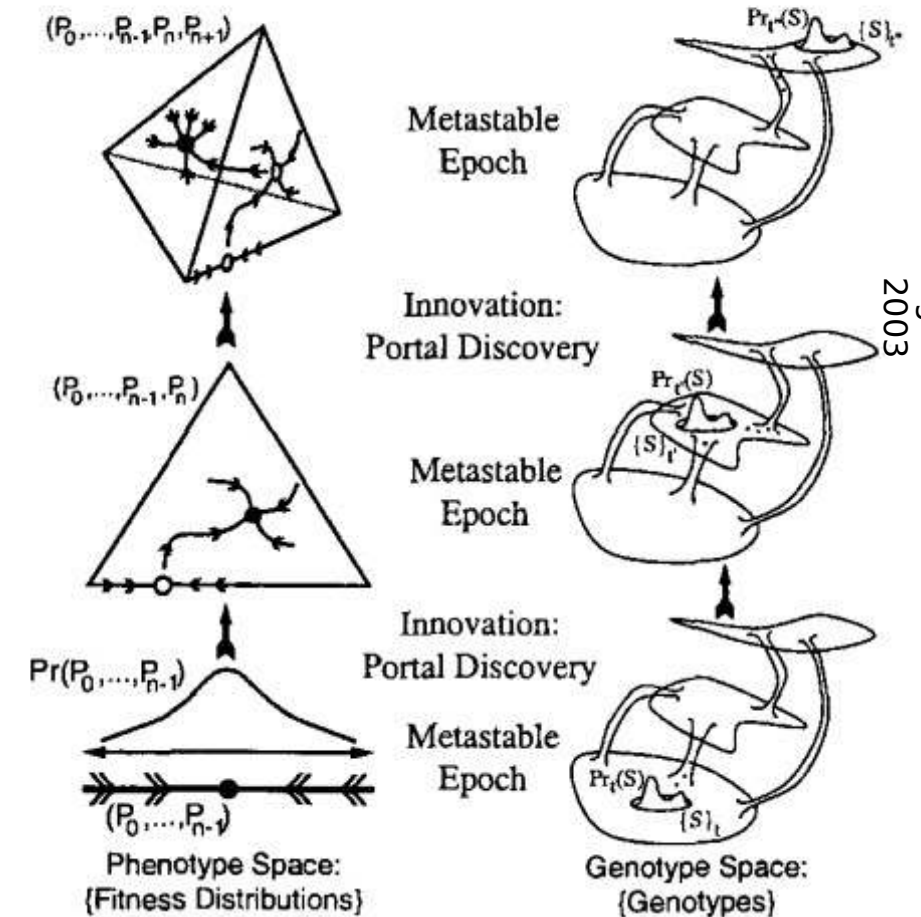
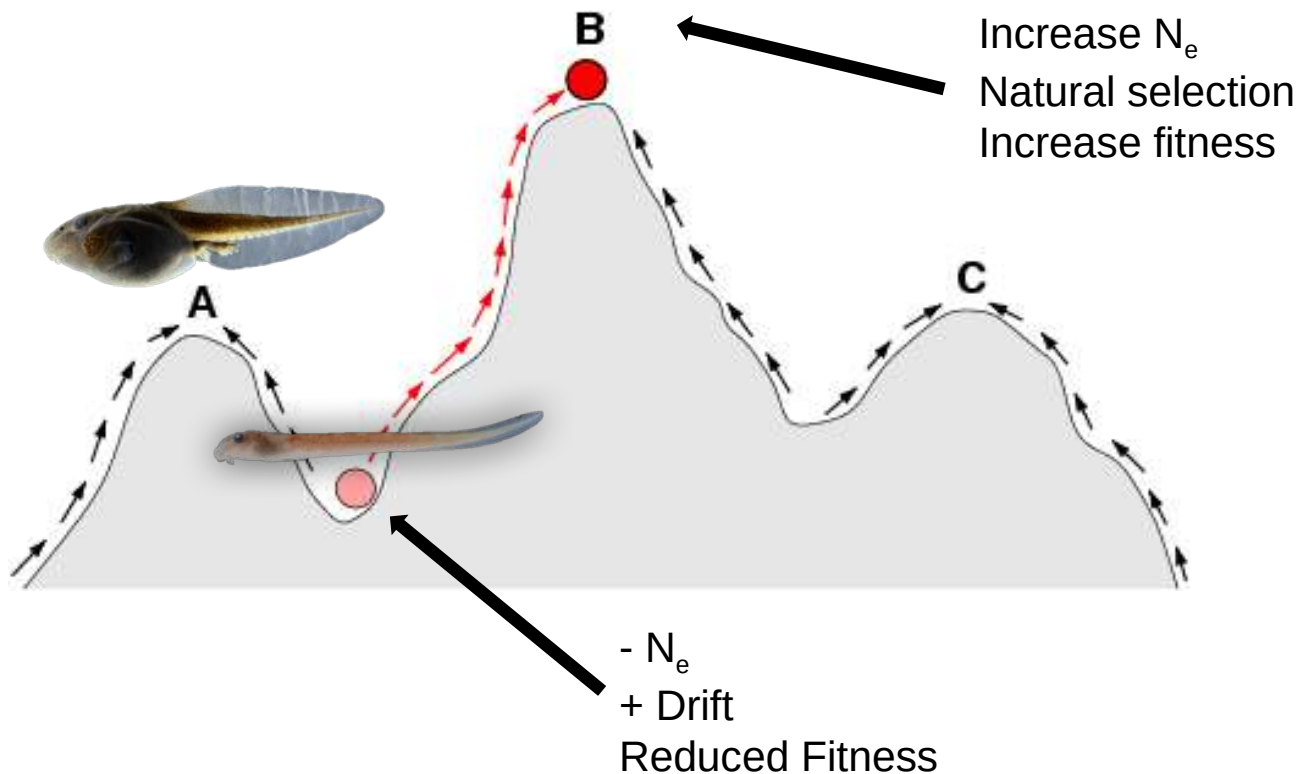
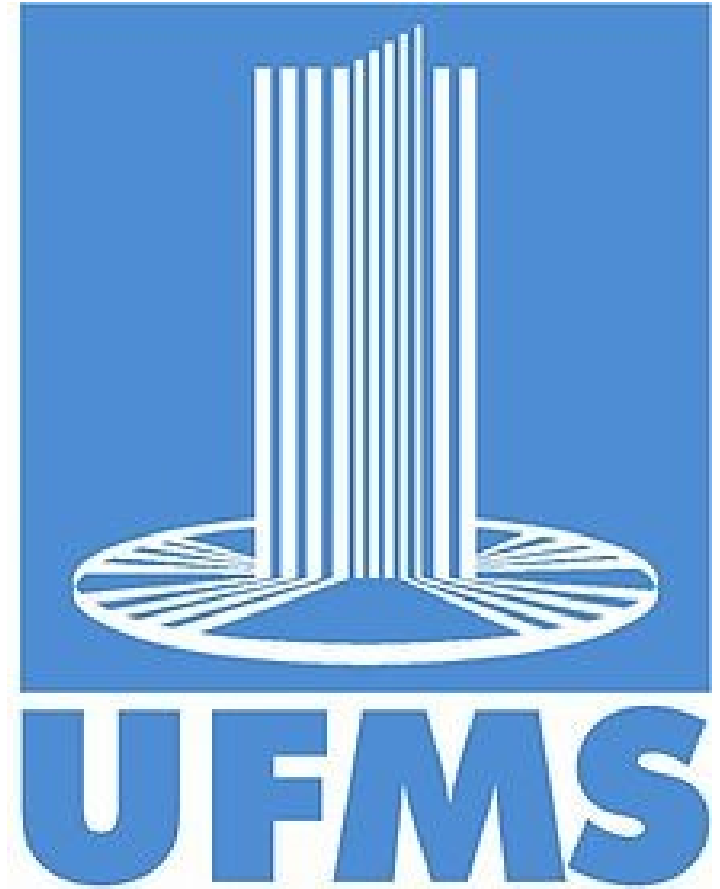


Fig. 8 Crutchfield 2003

Funding/Fellowship



All curators who provided access
to collections under their care



Questions?



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